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Who Gives Feedback Matters: Student Biases Towards Human and AI-Generated Formative Feedback

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ABSTRACT

Background: Feedback is essential for learning, helping individuals understand and improve their performance. However, providing timely, personalised feedback in higher education is challenging. Generative AI offers a scalable solution, yet little is known about students' biases towards AI-generated feedback.

Objectives: This study aims to investigate how the identity of the feedback provider (human vs. AI) affects students' perceptions of feedback quality and credibility.

Methods: The study involved 472 students across diverse academic programmes and levels in authentic educational environments and employed a within-subject experimental design with a priming effect. A mixed-methods approach combined quantitative analysis of feedback evaluations with qualitative insights into students' perceptions to deepen understanding of the observed biases.

Results and Conclusions: Students perceived AI as a significantly less credible feedback provider and tended to associate lower feedback quality with AI. Disclosing the feedback provider's identity led to decreased evaluations of AI-generated feedback and an increased preference for human-crafted feedback. These patterns were consistent across academic levels, genders, and fields of study. These insights highlight the need for targeted interventions, such as improving AI literacy and building human-in-the-loop systems, to mitigate biases and enhance the effectiveness of AI in educational feedback systems.

1 | Introduction

Feedback has always been a crucial part of the learning process. Properly formulated and timely feedback helps individuals understand and improve their performance (Hattie and Timperley 2007; Henderson et al. 2021; Wisniewski et al. 2020). In 2015, Hattie conducted a thorough synthesis of 1200 meta-analyses related to the influences of teaching activities and methods (e.g., traditional lectures, flipped classrooms, problem-based learning, using online and Internet-based solutions, etc.) on student learning and concluded that most likely 'the method of teaching is less critical than the attributes of the teaching within the methods' (Hattie 2015). In particular, this study strongly emphasised evidence-based teaching practises and the importance of feedback to enhance student learning outcomes. However, managing the feedback process is the most challenging part of teaching (Carless and Winstone 2023; Winstone and Carless 2019; Winstone et al. 2017). According to the 2023 Student Experience Survey (Australian Government Department of Education 2023), amongst all 'Teaching quality items,' the item aimed to evaluate the quality of feedback (formulated as 'Teachers commented on your work in ways that help you learn') got the lowest positive rates (< 64% of postgraduates and < 55% of undergraduate students) for five last years in a row. Similarly, according to the Annual Report and Accounts 2023–24 (The Office for Students 2023), the ratings for 'Assessment and Feedback' items were one of the lowest overall 'Teaching

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Lay Description

- What is already known about this topic
  - Advances in generative AI have opened up significant opportunities to provide scalable and high-quality formative feedback.
  - Users' evaluations of AI-generated content are significantly influenced by biases towards AI as a content provider.
  - These biases manifest in two opposing forms: algorithm aversion and algorithm appreciation.
- What this paper adds
  - We conducted an empirical study with 472 students in authentic learning environments to evaluate their perceptions of human and AI-generated feedback.
  - Students consistently attributed higher-quality feedback to human providers, even when the feedback was AI-generated.
  - Feedback explicitly identified as human-crafted was perceived as higher quality.
- Implications for practice and/or policy
  - It is essential to address student perceptual biases to maximise the effectiveness and acceptance of AI-generated feedback in education.
  - Enhancing students' AI literacy can help reduce negative biases and foster a more balanced evaluation of AI as a feedback provider.
  - Incorporating humans into the feedback loop by combining AI-generated feedback with human oversight or validation can enhance trust and ensure the quality of feedback.

and Assessment' items (< 78% of students). Indeed, providing meaningful, personalised feedback requires significant time and effort. This is especially true in higher education contexts, where personal attention from the instructors is limited due to large student cohorts. At the same time, students often come from diverse backgrounds and have varying levels of prior skills and knowledge (Buckingham Shum et al. 2023). As a result, the feedback delivery is often delayed, reducing its effectiveness, and the provided feedback is not specific to the concrete student performance (Williams 2024; Winstone and Boud 2022).

Automated feedback produced by AI can help speed up and scale the feedback delivery process. Generative AI, especially Large Language Models (LLM), is rapidly gaining momentum for automating feedback provision in educational contexts (Cavalcanti et al. 2021; Dai et al. 2023; Keuning et al. 2019; Lee et al. 2025; Meyer et al. 2024). Indeed, a large-scale user study ($N = 6960$) conducted in four major Australian universities confirmed that half of the participating students sought feedback from Generative AI as part of their learning routine (Henderson et al. 2025). Notable examples of LLM models used to provide feedback on student performance include a variety of fields and applications: in data science open-ended tasks (Dai et al. 2023), argumentation skills (Wambsganss et al. 2021), writing (Mejia-Domenzain et al. 2025; Escalante et al. 2023; Tang et al. 2024), source-based argument essays in history (Steiss et al. 2024), inquiry-based learning (Bewersdorff et al. 2023), algebra (Pardos and Bhandari 2023), and programming tasks (Pankiewicz and Baker 2023; Gabbay and Cohen 2024), achieving high performance in some cases even comparable to well-trained human evaluators.

However, most existing studies primarily target technological aspects like model accuracy and often miss the social–emotional aspects of AI's acceptance. As shown in other fields, the preference for human over AI advice, known as 'algorithm aversion' (Dietvorst et al. 2015), persists despite AI's proven efficiency. Although recent studies in education have started considering this issue, both from the educators' (Cukurova et al. 2020; Feldman-Maggor et al. 2025; Kaufmann and Budescu 2020; Kizilcec 2024; Nazaretsky, Cukurova, and Alexandron 2022; Nazaretsky et al. 2021; Williamson et al. 2025) and students' (Abendschein et al. 2024; Kim et al. 2020; Tsai et al. 2020; Radtke and Rummel 2025; Zeevy-Solovey 2024; Zou et al. 2025) perspectives, the field remains insufficiently explored, particularly concerning the psychological biases that might significantly shift practitioners' perceptions of AI-generated feedback (Abendschein et al. 2024; Kim et al. 2020; Tsai et al. 2020). In this study, we aim to reduce this gap by investigating to what extent students' knowledge of who provided the feedback—AI or a human—affects their evaluation of that feedback, with a particular focus on biases such as algorithm aversion and the tendency to favour human expertise.

2 | Related Work

2.1 | AI-Generated Formative Feedback Quality

A fast-growing body of research investigates the various LLMs' capacity to provide formative feedback at scale. Recent implementations have applied LLMs for feedback generation in writing (Almegren et al. 2024; Escalante et al. 2023; Mejia-Domenzain et al. 2025; Meyer et al. 2024; Steiss et al. 2024; Wambsganss et al. 2021), language learning (Zeevy-Solovey 2024; Zou et al. 2025), STEM subjects (Bewersdorff et al. 2023; Dai et al. 2023), and programming assessment (Heickal and Lan 2024; Gabbay and Cohen 2024; Pankiewicz and Baker 2023). Whilst several studies report that AI-generated feedback can match or even exceed the quality of human feedback in writing and STEM domains (Dai et al. 2023; Escalante et al. 2023; Fokides and Peristeraki 2025; Jacobsen and Weber 2023; Jansen et al. 2024; Jia et al. 2022; Steiss et al. 2024), others highlight serious limitations, including factual inaccuracies, instructional incoherence, and poor alignment with pedagogical principles (Stamper et al. 2024). These concerns have been noted across diverse contexts, such as reading comprehension (Seo et al. 2025), science education (Nguyen and Hayward 2025), argumentative writing (Yoon et al. 2023), and programming (Heickal and Lan 2024). In contrast to the aforementioned studies, in this paper, we do not aim to quantitatively assess the objective quality of AI-generated feedback or its comparison to human-crafted feedback. Instead, we focus

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on exploring how the knowledge of the feedback provider's identity affects students' perceptions of feedback.

2.2 | Motivational Aspects of Effective Formative Feedback

A considerable body of research exists in the area of design and implementation of effective formative human and AI feedback in the context of task-level performance (Buckingham Shum et al. 2023; Carless and Winstone 2023; Hattie and Timperley 2007; Narciss 2013; Shute 2008). This research advocates for a multi-dimensional approach to feedback, including feedback content and provider characteristics (Narciss 2013), and recommends a learner-centric approach to the feedback, considering the learners as active participants in the feedback process rather than passive receivers of the information (Boud and Molloy 2013; Lin et al. 2023; Wisniewski et al. 2020). To ensure the uptake of the feedback for effectively enhancing the learning process, feedback should be task-focused (Ryan et al. 2019; Shute 2008), provide objective performance evaluations (Boud and Molloy 2013), offer actionable advice on how to proceed (Ryan et al. 2019), and motivate learners to adopt it to improve their response (Shute 2008).

Based on the previous research, the following motivational factors influence learners' willingness to act upon feedback: learners' perceptions of feedback content objectivity (Roberson and Stewart 2006), usefulness (Henderson et al. 2021; Ryan et al. 2019), genuineness (Hirunyasiri et al. 2023; Shute 2008), and feedback provider credibility (Roberson and Stewart 2006). With respect to objectivity, according to the theories of procedural and informational justice, the feedback should be perceived by learners as fair and factual alignment between its content provided and the actual performance of the recipient (Boud and Molloy 2013; Roberson and Stewart 2006) and should be precise and specific to the response (Shute 2008; Van De Ridder et al. 2008). To act upon receiving feedback, it should be useful (Ryan et al. 2019), meaning informative, relevant, and applicable (Narciss 2013; Shute 2008). Applicability can be achieved by error-flagging and providing cues and hints on improving an answer whilst avoiding giving the correct solution itself (Narciss 2013). With respect to feedback genuineness, it should use a sincere, positive tone (Hattie and Timperley 2007) and be authentic, meaning honest and earned, not repeated often (Hirunyasiri et al. 2023) and focused on the learning process and not student abilities (Shute 2008). Concerning feedback provider credibility, the feedback source should be considered by the learners as safe and ethical (Tormey 2021). Safety refers to trustworthy, well-intentioned, and reliable feedback providers fostering confidence (McCroskey and Teven 1999; Tormey 2021), whilst ethics relates to qualities such as accountability, commitment to privacy, equality, and absence of biases (Tormey 2021).

In this research, we explored how students perceive the quality of AI and human feedback in terms of objectivity, usefulness, and genuineness, as well as their views on the competence of the feedback provider. These elements, informed by prior research, are shown to predict a student's inclination to apply the feedback constructively to improve their responses.

2.3 | Biases in Assessment of AI-Generated Content

Biases towards AI as a content provider can significantly influence users' evaluations of AI-generated content, manifesting in two opposing directions: algorithm aversion and algorithm appreciation (Dietvorst et al. 2015; Logg et al. 2019). Algorithm appreciation can result in over-reliance on AI due to overly optimistic expectations of its capabilities. This is often compounded by a lack of AI literacy, leading to automation bias, where users may assume that AI-generated feedback is always correct or more precise than humans (Nguyen 2024; Gómez et al. 2025). Supporting this, two studies by Ruwe and Mayweg-Paus (2023, 2024), involving 169 students and 98 student teachers, found that the AI feedback provider was perceived as more trustworthy than humans.

Conversely, algorithm aversion occurs when students are sceptical of AI's ability to deliver meaningful, accurate, or personalised feedback. This scepticism may arise from a lack of trust in AI systems, perceptions that AI cannot fully comprehend human learning contexts, or concerns about the impersonal nature of machine-generated responses (Nazaretsky, Mejia-Domenzain, et al. 2025; Kaufmann 2021). For example, Abendschein et al. (2024), using perception data collected from 155 undergraduate students, revealed that whilst the fictional AI feedback system was perceived as credible in terms of competence and trustworthiness, it was rated lower overall compared to the human grader delivering identical feedback. In addition, Henderson et al. (2025) found that although the vast majority of students (over 80% of 6960) perceived both AI- and human-generated feedback as helpful, there was a notable difference in the perceived trustworthiness: over 90% of students considered human feedback providers trustworthy, compared to only 60% for AI. In the context of essay writing, Zeevy-Solovey (2024) conducted a study with 30 higher education students to compare their perceptions of peer, AI, and teacher-provided feedback in an English as a Foreign Language (EFL) course. Their results showed that whilst students found both AI and human feedback equally helpful, they placed the highest trust in teacher feedback, viewing them as domain experts. Moreover, an additional study involving 20 EFL students revealed that learners engaged more actively with teacher-provided feedback than with AI-generated feedback, evidenced by a higher degree of integration of teacher comments into their revisions (Zou et al. 2025). In contrast, Radtke and Rummel (2025), who examined the essay revision behaviour of 303 participants, reported no significant differences in the number of identified areas for improvement or revision actions taken, regardless of whether the feedback was labelled as AI- or human-generated. Meanwhile, Williamson et al. (2025) observed in two experiments ($N = 570$) that educators favoured a complex knowledge tracing algorithm over a simpler, more transparent one within the ASSISTments platform—indicating a potential shift from algorithm aversion to algorithm appreciation, possibly influenced by the growing presence of Generative AI tools in education. These contradictory findings underscore the importance of investigating biases and trust in AI-generated feedback provision, as they play a pivotal role in determining how students engage with and benefit from AI as a learning tool, influencing both their reliance on and scepticism towards these systems.

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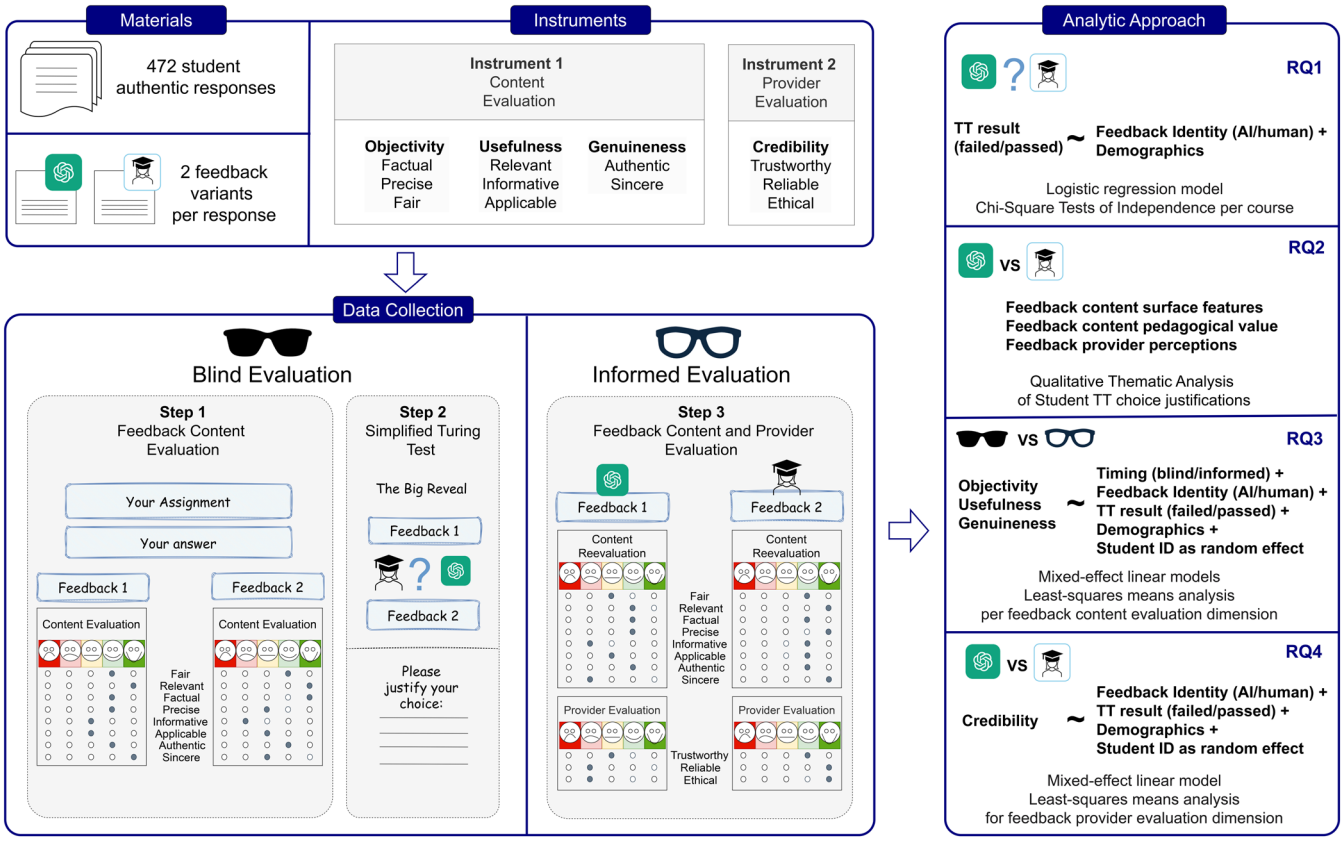

FIGURE 1 | The study pipeline includes materials, perception instruments, the experimental design of data collection, and an analytical approach for data analysis per research question.

3 | The Present Research

In this study, we aim to examine how students' awareness of the feedback provider's identity might influence their evaluation of feedback content, particularly in the context of algorithm aversion and preference for human expertise. With these goals in mind, the study seeks to address the following research questions: First, to what extent can students distinguish between AI-generated and human-crafted feedback (simplified Turing Test), and which factors influence their capacity to make this distinction (RQ1)? Second, which characteristics of the feedback do students use to differentiate between human and AI feedback providers (RQ2a), and to what extent is their judgement systematically biased (RQ2b)? Third, how do students' perceptions of the same feedback content change after the feedback provider's identity is revealed (RQ3)? Fourth, to what extent do students exhibit a negative bias towards AI as a feedback provider (RQ4)?

To achieve our goals, we conducted an extensive user study by analysing the responses of 472 higher education students from various courses, fields of study, and academic levels (Bachelor's and Master's). Specifically, we asked students to evaluate two feedback options (human-created and AI-generated) in their authentic learning settings twice: initially without (blind condition) and then with (informed condition) knowledge of the feedback provider's identity. This deliberate priming and reveal is designed to uncover changes in students' evaluative judgements, thus helping to evaluate the impact of the provider's identity on educational feedback perception and the impact of perceived feedback quality on perceived provider identity.

Our results reveal that students' ability to distinguish between AI-generated and human-created feedback depends on the underlying course and task. We also detect a change in feedback perception upon disclosure of the provider's identity, with a notable trend of increased favorability towards feedback identified as human-created and a decline in the evaluation of AI-generated feedback. Moreover, Turing Test outcomes are significantly correlated with feedback perception. In the blind condition, based on their subjective quality evaluation, most students attributed better feedback to humans. Consequently, students failing the test favour AI-generated feedback, whilst those passing prefer human feedback. We, therefore, can confirm a bias against AI as a feedback provider.

4 | Methodology

The study pipeline is presented in Figure 1. Below, we explain each experimental design component in more detail.

4.1 | Participants

Four-hundred and seventy two students from EPFL who were above the legal age participated in the study. Participation

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was voluntary. These students were enrolled in six different courses: Discrete Mathematics (ICC1 and ICC2), Information Theory (AICC), Applied Software Engineering (ASE), Applied Data Analysis (ADA), and Mathematical Analysis (MA). The courses span various academic programmes, including Computer Science, Communication Systems, Data Science, Civil, Mechanical, Life Sciences, Electrical and Electronic, Financial Engineering, and more. The data collection for this study occurred in fall 2023, and detailed socio-demographic information of the participants is presented in Table 1. Gender information in this study was collected using four predefined categories: male, female, non-binary, and other. Due to the very small number of participants identifying as non-binary ($n = 2$), conducting a separate statistical analysis for this group was not feasible, as such small subsample sizes lead to unreliable estimates and model instability. To ensure sufficient statistical power whilst retaining gender-diverse participants in the analysis, we grouped female and non-binary participants into a single category representing gender minorities. This approach aligns with methodological guidance for handling low-frequency identity categories in quantitative research whilst preserving inclusivity (Office of the Chief Statistician of the United States 2023). Analyses were thus conducted by comparing three groups: male, minority gender (female + non-binary), and other. The study was conducted with the approval of the Human Research Ethics Committee at EPFL (HREC 017-2023).

4.2 | Materials

We selected a pre-existing assignment for each course in the study and gathered genuine student responses. We adopted a positive formative feedback strategy, focusing on encouragement and support, recognising effort accurately, and highlighting improvement areas. We created two variants of feedback for each student, one from the teaching team and the other from AI. An illustration of human and AI-generated feedback, taken from a logical reasoning task in the AICC course, can be found in Figure 2. ICC1, ICC2, and MA courses were taught in French, and all other courses in English. More examples are presented in Supporting Information.

TABLE 1 | Demographics of participants by course and field of study, including degree, year, gender distribution (male/female/non-binary/other), and mean age (standard deviation).

Course	Field of study	Degree	Year	Participants	Age
ICC1	Mathematics, physics	BA	1st	65 (52/11/0/2)	19.1 (3.0)
ICC2	Civil & materials engineering	BA	1st	16 (9/7/0/0)	20.6 (4.8)
AICC	Computer science	BA	1st	187 (139/37/1/10)	19.3 (2.0)
MA	Mechanical engineering	BA	1st	28 (22/5/0/1)	18.9 (1.2)
ASE	Life sciences engineering	BA	2nd	70 (31/37/0/2)	20.3 (1.7)
ADA	Multiple	MA	1st	106 (72/31/1/2)	23.1 (2.9)

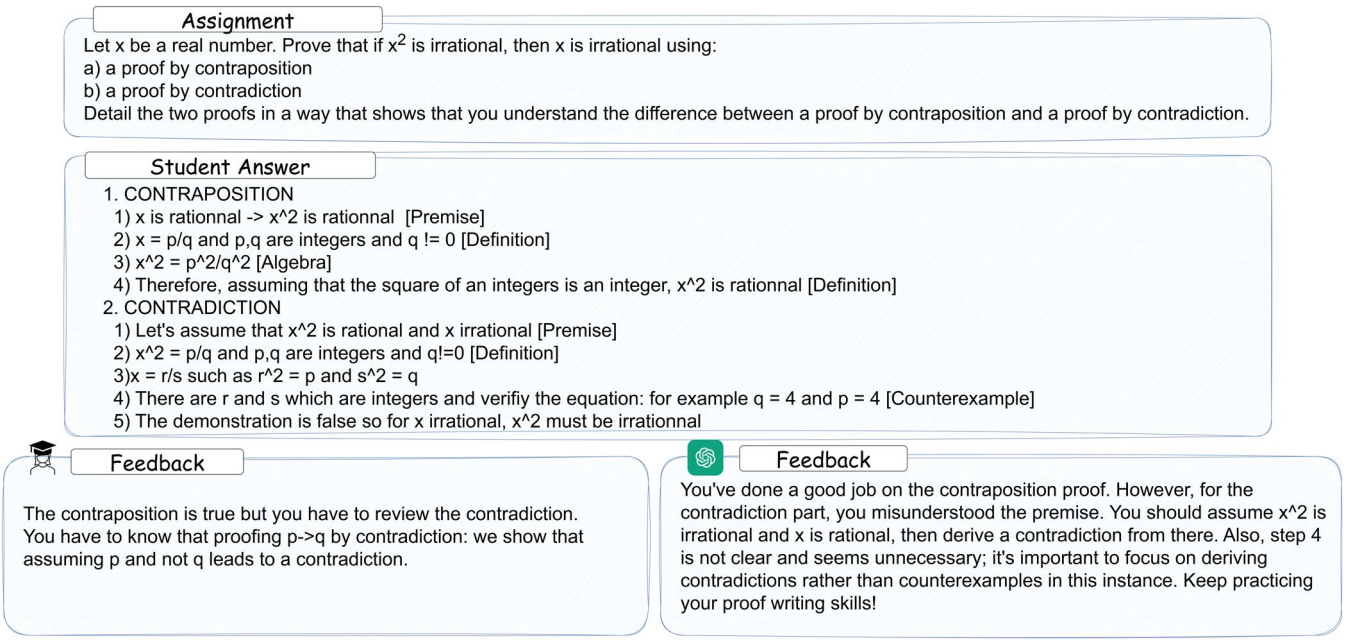

FIGURE 2 | An AICC course assignment, a corresponding student answer, and two variants of feedback: Human-created and AI-generated.

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4.2.1 | Human-Crafted Feedback

For each course, human feedback was provided by the original teaching teams. For MA, ASE, and ADA, existing authentic feedback was used as is, whilst in ICC1, ICC2, and AICC, formative feedback was not provided to the students as part of the routine. For these courses teaching teams were trained through a workshop on creating positive, formative feedback, emphasising feedback consistency and quality. The workshop included guidelines, feedback examples, and a communication channel for graders to ensure uniformity of feedback. More details about the types of assignments and number of human graders per course are presented in Table 2.

4.2.2 | Prompting Strategy for AI-Generated Feedback

Rather than evaluating the generative AI's ability to produce high-quality feedback, in this study, we focused on investigating students' qualitative evaluations of the provided feedback under blind conditions and when knowing the provider identity. To ensure valid inferences about students' perceptions and potential biases, the AI-generated feedback needed to be pedagogically and linguistically aligned with the human-crafted version. This comparability was crucial for minimising low-quality cues (e.g., LLM hallucinations) and preventing participants from identifying the feedback source based on superficial textual markers (e.g., an opening like 'Dear student'). To achieve this alignment, we adopted a targeted prompt engineering strategy designed to elicit responses from an LLM that plausibly mimicked the style, structure, and tone of authentic instructor feedback. Specifically, we used a *ROLE–CONTEXT–INSTRUCTION–FORMAT* prompting pattern,[1] which defined the LLM's role, the instructional context, the feedback type requirements, and the expected format of the output. Specifically, the prompt defined the *ROLE* of the AI as 'an excellent instructor', the *CONTEXT* including the course name, the assignment content, and the student's answer. Next, the *INSTRUCTION* was 'Please evaluate the student's answer and provide elaborated formative feedback,' followed by the format of the response. The final version of the prompt is presented in Figure 3. More information is presented in Supporting Information.

We refined the prompt through an iterative process until the generated feedback closely matched the quality and instructional style that students were used to receiving in their courses. Several principles guided our prompt design. For instance, we intentionally excluded the definition of elaborated formative feedback aligned with (Hattie and Timperley 2007) categories (e.g., Where to go?) from the prompt. When included, they caused the AI to produce unnaturally structured responses (e.g., using bullet points and explicitly naming rubric dimensions), which did not reflect the typical narrative style employed by instructors. Whilst a grading rubric existed for all the courses, we decided not to provide it in the prompt, as we wanted the LLM to focus not on the grades but on formative feedback. Conversely, instructors often concluded their feedback with a brief sentence summarising overall performance. We explicitly included this element in the AI prompt to replicate this authentic practise. The prompt results were iteratively evaluated by a teaching team of the AICC course till reaching a satisfactory final structure.

TABLE 2 | Overview of course assignments, human assessment routines, and additional feedback provision for various courses, detailing the number of human graders involved in the study.

Course	Assignment	Routine human assessment	Additional human feedback	Number of human graders
ICC1	C++ project	Grade	Code review	8
ICC2	Pseudo-code algorithm	Grade	Feedback	1
AICC	Logical reasoning proof	Grade	Feedback	9
MA	Logical reasoning proof	Feedback	—	1
ASE	Python project	Code review	—	17
ADA	Statistical inference task	Grade & Feedback	—	3

```
You are an excellent instructor teaching a course called [COURSE NAME].
You gave the students the following assignment: [ASSIGNMENT].
The student submission was [STUDENT ANSWER].
The correct solution is [SOLUTION].
Please evaluate the student's answer and provide elaborated formative feedback in [LANGUAGE].
Please follow the following instructions:
The feedback should be addressed directly to the student as is.
It should be no more than [NUMBER] lines.
Please provide one sentence of the overall evaluation at the end.
```

FIGURE 3 | Standardised prompt template used to create AI feedback for student assignments, detailing the necessary components and instructions.

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TABLE 3 | Feedback content and provider evaluation instruments.

Dimension	Terms	Supporting theory	References
Instrument 1: Feedback content evaluation. The question was formulated as follows: 'To what extent do you associate this feedback with the following term?'			
Objectivity	Precise, fair, factual	Theory of Procedural and Informational Justice (Roberson and Stewart 2006)	Boud and Molloy (2013); Shute (2008); Steiss et al. (2024)
Usefulness	Relevant, informative, applicable	Technology Acceptance Model (Davis 1989), Theory of Cognitive Load (Moreno 2004)	Hattie and Timperley (2007); Henderson et al. (2021); Narciss (2013); Ryan et al. (2019)
Genuineness	Authentic, sincere	Self-Determination Theory (Relatedness) (Ryan and Deci 2000)	Hirunyasiri et al. (2023); Abendschein et al. (2024)
Instrument 2: Feedback provider evaluation. The question was formulated as follows: 'To what extent do you associate this feedback provider with the following term?'			
Credibility	Trustworthy, reliable, ethical	Credibility Construct Measurement Scale (McCroskey and Teven 1999)	Abendschein et al. (2024); Tormey (2021)

We tested this generic prompt structure on the other courses by presenting the corresponding pairs of student responses and feedback examples to the course professors and teaching team. Based on their comments, some adjustments were made per course. More specifically, the correct solution was provided in the prompt for the ICC2, AICC, MA, and ADA courses only. We did not incorporate solutions to the coding projects in the ICC1 and ASE courses, as there was a wide range of perfectly correct answers, and our goal was to have the LLM focus on enhancing the students' responses instead of making comparisons to a single correct solution provided in the prompt. We also adjusted the output length to reflect common instructor practises: approximately five lines for short tasks in the ICC2, AICC, MA, and ADA courses and 10 lines for longer assignments such as coding projects in the ICC1 and ASE courses. We also adjusted the language of the feedback to reflect the language of the specific course being taught.

We used the GPT-4.0 Interface[2] for generating the AI feedback, as it was the most powerful LLM available at the time of data collection (Fall Semester 2023). Its advanced linguistic and reasoning capabilities enabled the generation of high-quality feedback that was sufficiently comparable to that written by experienced human instructors. This fidelity was key to isolating the effect of straightforward feedback attribution to a specific provider in the blind condition. To prevent hallucinations and contextually irrelevant content, the first author conducted a comprehensive review of all generated feedback variants. The GPT-generated feedback was provided to the students as is, except for removing the 'Dear student' greeting at the beginning and 'Your instructor' at the end (in some cases, GPT added them to the feedback text).

4.2.3 | AI-Generated and Human-Crafted Feedback Quality

The evaluation of the pedagogical quality of the corpus of AI-generated and human-crafted feedback used in this study is detailed in Nazaretsky et al. (2025), where we compare human and AI-generated feedback across four dimensions (Narciss 2013): (1) Red flags: incorrectness and low readability, (2) Cognitive: alignment with the student's current state of knowledge and advice on how to improve the response, (3) Metacognitive: the provision of advice for improving learning or self-regulated strategies, and (4) Motivational: effective praise. According to the statistical analysis, feedback quality was comparable between AI and human providers for the first two dimensions, whilst AI-generated feedback offered more metacognitive improvement suggestions and delivered praise more frequently (Nazaretsky et al. 2025).

4.3 | Perception Instruments

Following the framework proposed by (Narciss 2013; Wisniewski et al. 2020), we adopted a multidimensional approach to evaluating effective feedback by assessing feedback content (Instrument 1) and feedback provider characteristics (Instrument 2) as distinct components. Consistent with the methodology outlined in Tormey (2021), students' perceptions of both the feedback and the feedback provider were measured using a single question for each. Namely, [Instrument 1] 'To what extent do you associate this feedback with the following term?' and [Instrument 2] 'To what extent do you associate this feedback provider with the following term?' The students were asked to choose a Likert scale answer for each term. The Likert items were evaluated on a scale of five (Strongly Disagree = 0, Disagree = 1, Neutral = 2, Agree = 3, Strongly Agree = 4.)

We systematically derived the evaluation terms from prior research. More specifically, we identified three dimensions (*Objectivity*, *Usefulness*, *Genuineness*) and eight terms to evaluate the feedback content. In the context of the feedback provider, we identified one dimension: *Credibility*. A comprehensive overview of all dimensions, their associated terms, underlying psychological and pedagogical theories, and supporting references is provided in Table 3.

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4.4 | Data Collection

To evaluate how students' feedback perceptions depend on their knowledge of the feedback provider, we incorporated two aforementioned perception instruments into the data collection experimental procedure illustrated in Figure 1. Our study employed a within-subject experimental design with a priming effect aimed to blur the information about the feedback identity (Ragot et al. 2020). The study consisted of three steps (Data collection, Figure 1). During Step 1, we presented students with a routine assessment task from their course, their authentic answer, and two variants of feedback generated by human experts and the OpenAI GPT4 Interface (Section 4.2) and asked them to evaluate the feedback content. In Step 2, we revealed the existence of primer manipulation. Then, we asked students to guess the origin of each feedback (a simplified Turing Test based on one attempt only) and to explain and justify their choice through an open-ended question. In Step 3, we revealed the actual identity of each feedback provider and asked students if they wanted to change their initial opinions regarding the feedback content. In addition, we asked participants to evaluate the feedback providers. During the study, the students had no access to the correct solutions or their grades. We also collected students' demographics, such as their age and gender.

The responses to Step 2 (Figure 1, Turing Test) were used to answer RQ1 (the students' binary guess of feedback provider identity) and RQ2 (students' justifications and explanations about their choice). The Perception Instrument 1 responses (student blind and informed evaluations of feedback content quality) and Turing Test results were used to answer RQ3. In addition to the above, the data collected using Perception Instrument 2 (student evaluations of feedback provider credibility) were used to answer RQ4.

4.5 | Analytic Approach

For RQ1, RQ3, and RQ4 we employed quantitative analysis of the collected data, whilst RQ2 data (answers to an open-ended question) were analyzed qualitatively. Below, we present the methodology in more detail.

4.5.1 | Quantitative Analysis

First, to evaluate the reliability of the proposed instrument, we use Cronbach's α measure for each dimension *Objectivity*, *Usefulness*, *Genuineness*, and *Credibility* separately. Next, we calculate the scores of the dimension by averaging over all the terms associated with the dimension.

To study the possible correlation of demographics (age, gender, course) with students' ability to guess the feedback provider identity correctly (RQ1), we fit a Logistic Regression Model.[3] To analyse how the knowledge of the feedback provider identity influences students' perceptions towards feedback content quality (RQ3), we utilise a Mixed Linear Model (MLM)[4] approach. More specifically, we fit the MLM separately for each content dimension (*Objectivity*, *Usefulness*, and *Genuineness*) to analyse how the interaction between feedback provider (Human/AI), timing (Blind/Informed) and Turing Test results (TT Passed/TT Failed), and possible covariates (age, gender, course) influence the dimension scores. The individual variation between students is modelled as a random effect. The follow-up analysis of the means is done using the *emmeans* R package. Multiple comparisons are adjusted using the Tukey method. Finally, to explore how the feedback provider identity influences students' perceptions of the provider *Credibility* (RQ4), we fit a Mixed Linear Model[4]. More specifically, we model an interaction between feedback provider *Credibility* and feedback provider identity (Human/AI), along with possible covariates (course, age, gender, and TT results), whilst representing the individual variation between students as a random effect.

All survey questions were defined as mandatory, so there were no missing data points. Independence of observations was assumed based on the study design. For RQ3 and RQ4, where multiple responses were nested within individual students (Human/AI, Blind/Informed conditions), we included a random intercept for student ID to account for within-subject dependence. Potential multicollinearity was examined in all linear models using the variance inflation factor (VIF)[5] for each predictor variable. All VIF values were below the commonly accepted threshold of 3, indicating no problematic multicollinearity amongst the predictors (Hair Jr et al. 2010). To account for potential heteroskedasticity, we used robust standard errors in all linear regression models. For the χ^2 tests, we confirmed that the expected cell counts met the minimum threshold (> 5), ensuring the validity of the test.

4.5.2 | Qualitative Analysis

We used a combination of a top-down and bottom-up approach to analyze students' responses to the open-ended question (Figure 1, Step 2). Each response was considered a unit of analysis. Although the study was conducted in English, some students ($n = 22$) preferred to respond to the open-ended question in French. Their responses were automatically translated into English (using the Google Translate tool[6]), and the correctness of the translation was verified by the fourth author, a native French speaker. The analysis was done in an iterative manner. First, the first author carefully read all the responses to extract the characteristics (mainly represented in students' responses as adjectives, e.g., precise, factual, helpful, etc.) used by students to explain and justify their Turing Test decisions. Second, first author assigned these characteristics to the top-down categories. The categories were informed by the previous research and included the following pedagogical qualities (Section 2.2): (1) feedback objectivity; (2) feedback usefulness; (3) feedback genuineness, as well as three 'surface' factors: (4) structure and grammar of feedback text; (5) feedback text readability and clarity; and (6) feedback text awareness of broader learning context (see Table A1 for more details). Next, the first and second authors analyzed the feedback characteristics mentioned by the students and came to 100% agreement about which category it belongs to. During the discussions,

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we also decided to divide some of the categories into subcategories as specified in Table A1. We constructed a detailed labelling scheme that included each adjective or other characteristic used in student responses (even once only). It is essential to mention that students used both positive and negative sentiment to characterise the same features (e.g., kind-rude, friendly-cold, etc.) Finally, the first author used this scheme to label the responses. For each response, the labelling included marking the categories mentioned to describe the feedback *perceived*(!) as human-crafted and AI-generated separately, including labelling the corresponding sentiment. It is important to mention that at the time of evaluation, the students did not know the identity of the feedback provider. This means that those who failed TT evaluated human-crafted feedback erroneously, thinking it was AI-generated and vice versa.

The categories, subcategories, and descriptions are summarised in Table A1. In the results section below, we elaborate on the emerged categories and subcategories and provide examples of the student responses and their labelling. More examples of student responses are presented in Table A2.

5 | Results

We analysed the students' answers to the blind evaluation, the TT, and their informed evaluations to investigate whether students can distinguish between human and AI feedback (RQ1), how students distinguish between AI-generated and human-crafted feedback, and whether their judgement is biased (RQ2a and RQ2b). We also examined how their perception changes when the feedback provider's identity is revealed (RQ3) and whether they hold biases against AI as a feedback provider (RQ4).

The internal consistency of the proposed instruments (measured by Cronbach's α) was good ($\alpha \geq 0.80$) for the *Objectivity* ($\alpha = 0.82$), *Usefulness* ($\alpha = 0.87$), and *Credibility* ($\alpha = 0.85$) dimensions and excellent ($\alpha \geq 0.90$) for the *Genuineness* ($\alpha = 0.91$) dimension. Next, we calculated the scores of the dimensions by averaging over all terms associated with the dimension over the responses of the 472 students.

5.1 | Turing Test for AI and Human Feedback (RQ1)

In total, 287 out of 472 students could guess the feedback provider correctly. To study the possible correlation of demographics (age, gender, course) with students' ability to guess the feedback provider's identity correctly, we fit a Logistic Regression Model. The Logistic Regression model revealed that students' ability to identify feedback provider identity correctly was not correlated significantly with their age and gender. However, there was a significant correlation with course topics. Given the relatively small sample sizes in most courses ($n < 100$), Binomial Tests were conducted to assess whether students correctly identified the feedback provider at rates significantly different from chance (50%) within each course. A Bonferroni-adjusted alpha level of $p = 0.05/6 = 0.0083$ was applied to account for multiple comparisons across the six courses. The results showed that students in ICC1 ($p < 0.001$, $n = 65$, 79% correct responses) and ASE ($p < 0.001$, $n = 70$, 80% correct responses) identified the provider at rates significantly above chance. No other course reached statistical significance after correction, including ICC2 ($p = 0.021 > 0.0083$, $n = 16$, 81% correct responses, not significant after Bonferroni correction), AICC ($p = 0.306$, $n = 187$, 54% correct responses), MA ($p = 0.087$, $n = 28$, 68% correct responses), and ADA ($p = 0.206$, $n = 106$, 43% correct responses), suggesting that students in those courses were not reliably able to distinguish between AI-generated and human-crafted feedback.

5.2 | Feedback Characteristics Students Employed to Distinguish Between Human and AI Feedback Providers (RQ2a)

The Turing Test open-ended question (Figure 1, Step 2) was defined as mandatory. However, not all the students wrote informative responses. More specifically, we removed from the analysis all the answers that included not meaningful character patterns (e.g., '...' or 'Parce que') and answers that did not provide meaningful information about feedback text evaluation (e.g., 'This is pure guessing, so out of intuition', 'Gut feeling' or 'Instinct'). In total, 25 responses were excluded, and a qualitative analysis was performed on the remaining data ($n = 447$). The response lengths varied between 2 and 205 words, with a median length of 20 ($\mu = 28$, $\sigma = 28$). The number of feedback features mentioned in the student responses varied between 1 and 12, with a median of 2 ($\mu = 2.9$, $\sigma = 3.2$), with 289 students mentioning at least two different feedback dimensions. Ninety-three students relied only on surface features when justifying their choice, 191 only on pedagogical quality, and 163 on both.

Below, we provide analyses of students' perceptions of AI-generated and human-crafted feedback in more detail, based on the features and categories grounded in the labelling scheme presented in Table A1 (see also Table A2 for examples of student responses per labelling category). We analyze each feature's discriminative power, defined as the extent to which it was consistently (or with rare exceptions) attributed exclusively to humans or AI.

In addition, Table 4 provides a comprehensive summary of the RQ2a findings by reporting the frequency with which students mentioned each category and subcategory of Turing Test justifications, classified by sentiment (positive or negative), along with the percentage of mentions attributed to AI. This table focuses on the total number of mentions rather than the number of individual students referencing each feature. This distinction is important, as students evaluated two feedback variants and could mention the same feature multiple times, for example, positively when describing the human-crafted feedback and negatively when evaluating the AI-generated feedback for the same response.

5.2.1 | Surface Features Analysis

In total, 250 (56%) out of 447 students considered surface features. Amongst the surface features, the most discriminating were Structure & Grammar and knowing the Broader Learning Context. One-hundred and fourteen students mentioned

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TABLE 4 | Number of times students mentioned each category and subcategory of turing test justifications, classified by sentiment (positive or negative), including the percentage attributed to AI for each case.

Category	Description	# of Mentioned in positive tone (% attributed to AI)	# of Mentioned in negative tone (% attributed to AI)
Surface features			
Structure & grammar	Formatting consistency, flawless grammar	91 (100%)	34 (0%)
Clarity & readability	Concise/easy to understand	41 (54%)	65 (92%)
Broader learning context	Comments about learning and teaching routine habits, and pedagogical context	49 (0%)	24 (100%)
	Comments about pedagogical team shortage in time/ resources/enthusiasm	—	9 (0%)
Pedagogical features			
Objectivity	Precise	83 (20%)	73 (88%)
	Factual	14 (86%)	—
	Personalised	28 (4%)	28 (96%)
	Correct/accurate	24 (4%)	64 (95%)
Usefulness	Helpful	56 (18%)	93 (98%)
	Reflects on what was right in the answer	10 (70%)	3 (100%)
	Reflects on what was wrong in the answer	29 (40%)	18 (94%)
	Provides concrete advice/hint	35 (18%)	31 (100%)
	Provides strategic advice	5 (20%)	1 (100%)
	Provides any new information	—	18 (100%)
Genuineness	Human tone, empathy, encouragement	46 (6%)	64 (97%)
	Generic encouraging phrases	—	35 (67%)
	Over-positive and over-praising	—	45 (100%)

Note: To illustrate, students mentioned 83 times that the feedback variant was precise. Amongst them, only 20% attributed such feedback to the AI feedback provider. In addition, students mentioned 73 times that the specific feedback variant was not precise. In this case, 88% of them attributed this feedback variant to AI.

features related to the Structure & Grammar of at least one of the two feedback variants they evaluated. Feedback perceived as well-formatted, grammatically error-free, and written in formal language ($n = 91$) was consistently, with no exceptions, attributed to AI. Conversely, texts containing typos, emojis, and poor formatting ($n = 34$) were always considered to be written by humans. Seventy-three students mentioned familiarity ($n = 49$) or unfamiliarity ($n = 24$) with the broader learning context as a factor guiding their judgement about feedback sources. Ninety-six students (with 106 comments) used the Clarity & Readability feature to base their TT decision. Their expectations varied, making this feature less discriminative. Whilst some students ($n = 70$) assumed that AI feedback would be vague, verbose, ambiguous, and less concise and coherent than human feedback, others ($n = 26$) assumed the opposite. Interestingly, some students ($n = 9$) believed the pedagogical team lacked the time and enthusiasm to provide high-quality formative feedback (e.g., 'I don't think assistants have that much time to write such detailed and elaborate feedback for every student').

5.2.2 | Pedagogical Features Analysis

Out of 447 students, 355 (79.4%) considered pedagogical feedback features, whilst 197 (44%) relied solely on these features to make their judgements.

5.2.2.1 | Objectivity. Feedback objectivity emerged as the most crucial factor for students when making their TT decisions. Indeed, 177 students mentioned characteristics of this category in their justifications for their TT choices. More specifically, 95 students (with 156 comments) evaluated the precision of the feedback variants, with only 18 attributing the less precise feedback to humans. Moreover, accuracy emerged as

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another crucial factor for 64 students (with 88 comments), serving as a key criterion for distinguishing between AI and human feedback providers. Indeed, in case students ($n = 64$) thought the feedback was inaccurate (e.g., it did not identify all the errors they had in their response or did not understand what was written in their response correctly), they almost always attributed it to AI ($n = 61$). Only three students assumed that the pedagogical team could produce inaccurate feedback. Moreover, 35 students (with 56 comments) argued about how much feedback was specific to their response, with 34 attributing (what they perceived as) more personal and tailored feedback to the human feedback provider and only one to AI. At the same time, students who mentioned the extent to which feedback was factual ($n = 14$) considered the AI feedback provider a source of more factual and objective feedback ($n = 12$).

5.2.2.2 | Usefulness. Feedback usefulness was essential, with 171 students using it as a parameter in their Turing Test decisions. One-hundred and seven amongst them (with 149 comments) made rather general comments about feedback helpfulness, applicability, and relevance, with 97 attributing these positive features of formative feedback to human feedback providers and only 10 to AI. Ninety-five students (with 150 comments) provided more specific explanations on whether and how the feedback reflected their current knowledge and provided advice on improving it. Fifty-eight students considered to what extent the feedback pointed out what was right ($n = 13$) and wrong in their answers ($n = 43$) or none of the above ($n = 18$). In most cases ($n = 32$), if the feedback provided useful information about students' errors, they attributed it to humans. In all the instances where students ($n = 18$) mentioned that feedback only reformulated what they did without giving any useful information on the correctness of the answer, they attributed it to AI. Fifty-six students explicitly mentioned whether the feedback provided advice, a hint, or a method for improving, with 50 attributing this critical feature of elaborated feedback to human-crafted feedback and only 6 to AI. Notably, only 6 students mentioned in their TT justifications that feedback provided advice on improving their learning process (e.g., 'AI feedback describes more what to do than explaining about how to use and understand the method.') Amongst them, only one attributed this positive feature to AI.

5.2.2.3 | Genuineness. Ninety-nine students (with 110 comments) mentioned communication style and tone as an additional distinguishing feature of human feedback. In almost all cases ($n = 94$), the genuine and warm communication style was attributed to humans, whilst cold, distant, and 'inhuman' was attributed to AI. Only five students attributed more friendly and authentic feedback texts to AI. The students also related to encouraging tone and praise ($n = 80$), with 12 attributing effective praise to humans and 23 to AI. In the other cases ($n = 45$), students perceived feedback as overly praising, overly optimistic, or not earned and consistently attributed it to the AI feedback provider.

Notably, the same generic encouraging phrases, such as 'Good job!', 'Keep practicing!' and 'Keep up the good work!' ($n = 23$) were attributed to humans ($n = 9$) and AI ($n = 14$) with no clear preference. However, students reported experiencing different sentiments depending on the feedback source. When they assumed these phrases came from AI, they, in most cases, perceived them as unnatural, strange, or even intimidating. One student expressed their feelings, '"Keep practicing!", no, I'll just stay and look at the floor.' In contrast, when the exact phrases were attributed to humans, they were always seen as genuinely supportive. One student noted, 'In human feedback, there are more encouraging words to make us happy, and I think that is part of teaching.'

5.3 | Exploration of Biased Judgements (RQ2b)

To further explore possible biases, we examined instances where students' perceptions of the AI feedback provider led to misjudgments and incorrect identification of the provider, resulting in a failed TT. Amongst the students who used surface features (solely or in conjunction with pedagogical features) to justify their TT decision ($n = 250$), only 84 failed TT, indicating ($\chi^2 = 26.9, df = 1, p < .001$) that surface features were a reliable hint to distinguish between AI and human feedback providers. Whilst amongst the students who mentioned only pedagogical features ($n = 197$), 90 failed TT, indicating ($\chi^2 = 1.467, df = 1, p = .23 > .05$) that employing them only led to a near-random guess about feedback provider identity. To explore this phenomenon in more detail, we investigated the number of instances of feedback considered of low quality, such as incorrect, imprecise, impersonal, etc., in relation to actual and perceived feedback providers (Figure 4). The results confirmed a tendency to disproportionately attribute feedback to AI, based on its perceived low pedagogical quality, even when a pedagogical team actually provided it.

In summary, the analysis revealed a strong consensus amongst students that AI-generated feedback was perceived as less objective (characterised as imprecise, inaccurate, and impersonal), less useful (unhelpful, lacking evaluation of current knowledge state, and lacking elaboration on improvement suggestions), and less genuine (overly formal and over-praising). At the same time, human feedback was primarily characterised using positive adjectives. In contrast, an objective comparison of feedback quality between AI and human feedback providers (Nazaretsky et al. 2025) indicates that their quality was practically equivalent. This discrepancy suggests that students exhibited biased perceptions of AI feedback, particularly concerning the pedagogical features mentioned earlier.

5.4 | Impact of Identity Disclosure on Perceptions (RQ3)

To analyse how the knowledge of the feedback provider identity influences students' perceptions towards feedback content quality, we utilised a Mixed Linear Model (MLM)[4] approach. More specifically, we fit the MLM separately for each content dimension (*Objectivity*, *Usefulness*, and *Genuineness*) to analyse how the interaction between feedback provider (Human/AI), timing (Blind/Informed) and Turing Test results (TT Passed/TT Failed), and possible covariates (course, age, gender) influence the dimension scores. The individual variation between students was modelled as a random effect. The analysis of the

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MLM models detected no effect of course, age, or gender on the students' perceptions across all dimensions; however, there was a significant effect ($p < 0.001$) with feedback providers.

Interestingly, in the follow-up analysis of the means, distinct patterns emerged between perceptions towards AI-generated and human feedback. Students tended to increase their scores for the feedback provided by human evaluators after being informed about the feedback provider's identity. However, the change was not statistically significant in all the content evaluation dimensions (Figure 5a, Human condition). At the same time, for AI feedback, the students tended to decrease their

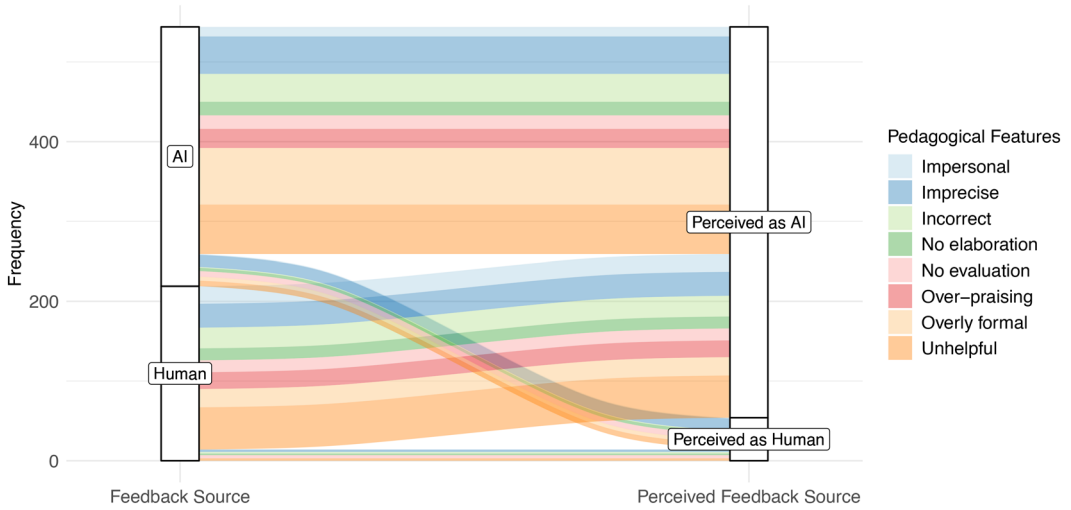

FIGURE 4 | The flow between the actual feedback source (AI or Human) and the perceived feedback source (Perceived as AI or Perceived as Human) across various negative pedagogical feedback features as identified by students. The x-axis shows the actual source of feedback (left) and the perceived source (right), whilst the y-axis represents the frequency of instances. Each coloured flow corresponds to a specific pedagogical feature (e.g., 'Imprecise,' 'Incorrect'). The width of each flow indicates the frequency of instances where feedback was perceived as having the feature. In the case of AI feedback ($n = 325$), only 40 perceived low-quality feedback variants were incorrectly attributed to humans. For the feedback provided by humans ($n = 219$), 205 instances were misperceived as AI, and only 14 were correctly identified as human.

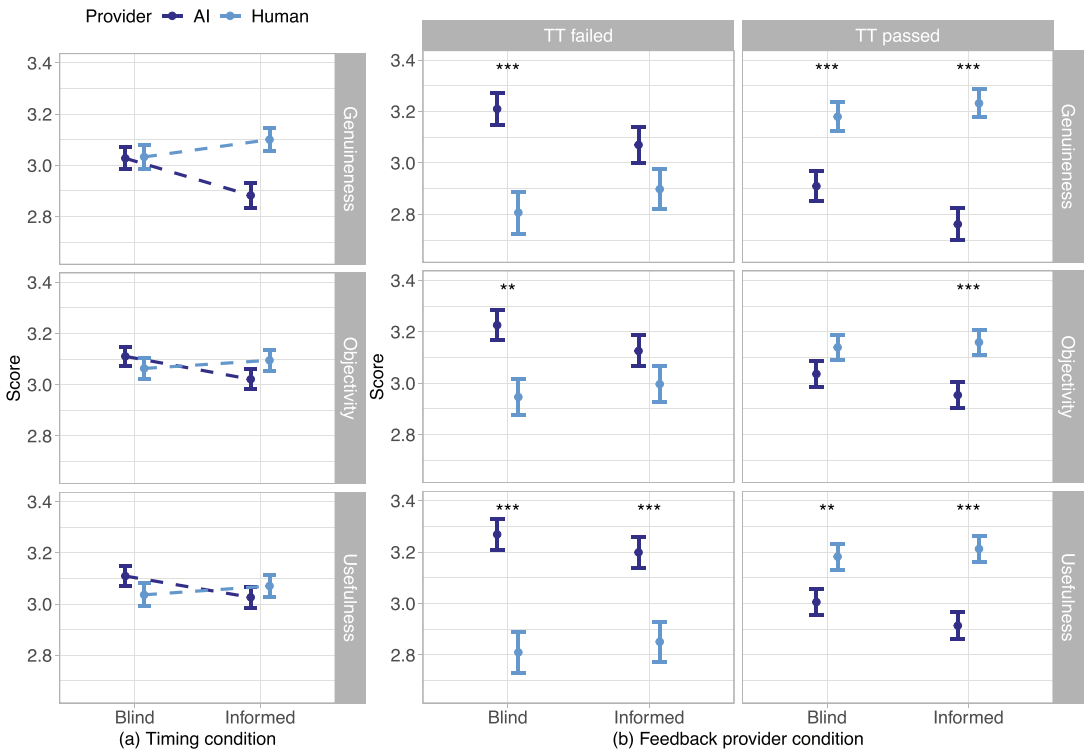

FIGURE 5 | Student perceptions towards feedback content per dimension (*Genuineness*, *Objectivity*, *Usefulness*) by interactions with Feedback provider (Human/AI), Timing (Blind/Informed), and Turing Test results (TT failed/TT passed). Scores: Strongly disagree = 0, Disagree = 1, Neutral = 2, Agree = 3, Strongly agree = 4. Significance codes: ***< 0.001, **< 0.01, *< 0.05.

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scores in all the categories evaluating feedback content after being informed that AI generated it. The change was statistically significant for the *Genuineness* dimension ($p < .01$), but not for the *Objectivity* and *Usefulness* dimensions (Figure 5a, AI condition). These results did not correlate with the Turing Test results.

Subsequently, we analyzed the differences between the content evaluation of the human-created and AI-generated feedback for both timing conditions (Blind/Informed) separately. The MLM analysis indicated a correlation with the Turing Test results in this case. Based on the follow-up analysis of differences in the means, we found distinct patterns that emerged between perceptions towards AI-generated feedback and human feedback for students who succeeded and failed to guess the feedback provider's identity correctly (TT passed/TT failed). More specifically, in blind conditions, students who failed the TT evaluated AI feedback significantly higher than human feedback amongst all dimensions. However, after being informed about the provider identity, they reduced their scores for AI and increased their scores for human feedback, such that the difference was not significant anymore for the *Genuineness* and *Objectivity* dimensions (Figure 5b, TT failed condition). Conversely, in blind conditions, students who passed the TT rated AI feedback lower than human feedback. However, the difference was significant for the *Genuineness* dimension only. After being informed about the provider identity, they reduced their scores for AI and increased their scores for human feedback, such that the difference became significant in all dimensions (Figure 5b, TT passed condition).

5.5 | Bias Against AI as a Feedback Provider (RQ4)

To explore how the feedback provider identity influences students' perceptions of the provider *Credibility*, we fit a mixed linear model[4]. More specifically, we modelled an interaction between feedback provider *Credibility* and feedback provider identity (Human/AI), along with possible covariates (course, age, gender, and TT results), whilst representing the individual variation between students as a random effect. The results of MLM revealed a significant positive correlation of human identity with feedback provider *Credibility* ($p < .001$) and a negative correlation with participants' age ($p < .01$). We found no significant correlation with the course, gender, and ability to guess the feedback provider identity. The results are presented in Figure 6a. The follow-up Wilcoxon rank sum test indicated a medium effect (Cohen's $d = 0.57$) of the feedback provider identity on feedback provider *Credibility*, indicating that feedback from human providers ($\mu = 3.28, \sigma = 0.77$) is perceived as significantly more credible than feedback generated by AI ($\mu = 2.25, \sigma = 0.85$). In addition, we used Wilcoxon rank sum tests to analyse if the perceived feedback quality (in blind condition) differs amongst perceived feedback provider identities (as identified by student TT choices) per feedback content quality dimension. The results are presented in Figure 6b. The difference was significant for all the dimensions. However, the effect size was small ($d < 0.2$).

6 | Discussion and Implications

AI offers the capability to deliver personalised, timely feedback at scale. As a result, research on this topic has grown, primarily emphasising feedback quality whilst often overlooking the social–emotional dimensions of AI adoption (Cavalcanti et al. 2021). Undoubtedly, feedback quality is an important factor in students' perceptions of AI-generated feedback. Concerning our dataset, we confirmed that the pedagogical quality of the AI-generated feedback was on par with the human-crafted feedback (Correctness and Cognitive aspects), and even outperformed it in other aspects (Metacognitive and Motivational) (Nazaretsky et al. 2025). Reflecting this, in the blind condition, students' perceptions of feedback quality over all feedback content dimensions (*Objectivity*, *Usefulness*, and *Genuineness*) were similar for AI and human feedback providers (RQ3, Figure 5a). At the same time, students' feedback quality perceptions of *what they thought* was human-crafted feedback, were significantly higher in all the dimensions (RQ4, Figure 6b). These results emphasise that some

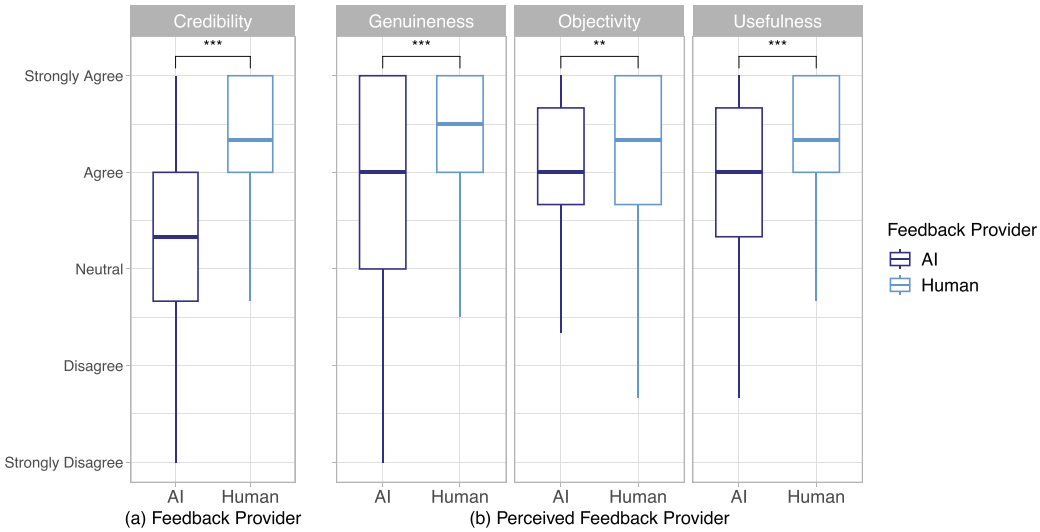

FIGURE 6 | Bias against AI as a feedback provider. (a) Student perceptions towards feedback provider *credibility* in informed condition. (b) Student perceptions towards feedback content quality over *genuineness, objectivity, usefulness* dimensions in blind condition. Scores: Strongly disagree = 0, Disagree = 1, Neutral = 2, Agree = 3, Strongly agree = 4. Significance codes: ***< 0.001, **< 0.01.

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factors influencing students' perceptions of AI-generated feedback are not directly related to the feedback content quality but to the social–emotional aspects of AI adoption. To explain this gap, our study investigated students' perceptions of AI-generated and human-crafted feedback, focusing on discovering potential biases against AI as a feedback provider. A significant observation was the reciprocal relationship between perceived feedback quality and the perceived identity of the provider.

First, students attributed higher-quality feedback (based on subjective evaluations in the blind condition, RQ2b) to human providers. Indeed, the majority of positive pedagogical features (e.g., being 'precise', 'personalised', 'accurate', and 'useful') were attributed to human feedback providers. In contrast, the lack of these features was systematically attributed to the AI feedback provider (Table 4). The only positive pedagogical feature attributed primarily to AI was 'factual'. However, factual correctness alone may not be sufficient to address the broader emotional and pedagogical needs that students expect from high-quality formative feedback. These results may stem from the students' disbelief that models like GPT could produce human-like, high-pedagogical-quality feedback, which reinforced their assumption that such feedback came from the teaching team, ultimately leading to failing the Turing Test.

Second, based on RQ3 results, students decreased their evaluation of the same feedback content after being informed it was AI-generated. The decrease was significant for the *Genuineness* dimension, measured by *Authentic* and *Sincere* terms, and can be explained by evaluating qualities traditionally attributed to humans. However, the trend of decreased evaluations in the *Objectivity* and *Usefulness* dimensions is more complex to interpret. Based on previous research, students, as experts in training, often are not confident in their objective evaluation of educational content (Price et al. 2010). In such situations of uncertainty, human evaluation of objectivity and usefulness of the information depends mainly on their trust in the information provider (Nazaretsky, Ariely, et al. 2022). When students were informed about the identity of the feedback provider, they adjusted their evaluations according to their confidence in the feedback provider. Indeed, once the true AI identity of the feedback was disclosed, their lower trust in AI as a feedback provider (based on RQ4 results) influenced their feedback assessment, leading to reduced scores in these dimensions and vice versa for the human-crafted feedback.

Third, based on RQ4 results, the human feedback provider was consistently viewed as more credible, with higher-quality feedback frequently attributed to human sources. These findings are on par with (Kaufmann 2021; Abendschein et al. 2024; Nazaretsky et al. 2021), however opposite results were presented in Ruwe and Mayweg-Paus (2024, 2023). These contrasting findings can be explained by the simultaneous presence of algorithm aversion and algorithm appreciation (Jussupow et al. 2024), reflecting the dual nature of students' perceptions towards AI systems. Whilst students may value the efficiency and objectivity of AI-generated feedback for routine tasks, they may remain sceptical about its ability to address complex, nuanced, or highly personalised educational needs.

6.1 | Recommendations for Educational Practitioners

6.1.1 | Addressing Lack of Transparency and Ethics Concerns

Echoing the findings of Abendschein et al. (2024); Kaufmann (2021); Nazaretsky, Cukurova, and Alexandron (2022), our study reveals a strong preference for human guidance over AI-generated suggestions (based on RQ2 and RQ4 findings), indicating a fundamental human inclination for personal interaction and judgement. In this light, AI-powered system designers might be tempted to conceal or blur the AI nature of feedback providers, justifying this by the system's ability to deliver high-quality feedback. However, we contend that such practises are unethical. Transparency regarding AI involvement in feedback generation is crucial. One reason is the variability in feedback quality between controlled test environments and actual educational settings (Yin et al. 2019). Another reason is the educational value of teaching students to recognise and handle potential AI errors (Bearman et al. 2024; Nguyen 2024). Educating students to critically assess feedback from accessible but imperfect AI technologies like ChatGPT prepares them to continue leveraging such tools for lifelong learning and their future professional environments, enhancing their critical thinking and self-regulated learning skills.

6.1.2 | Promoting Student AI and Feedback Literacy

Approaching feedback quality from a different perspective, it is notable that only 6 (<%) students explicitly mentioned feedback features aimed at improving their critical thinking and learning strategies (i.e., metacognitive feedback aspect (Narciss 2013)). In contrast, a considerably larger proportion of students (around 20%) focused on whether the feedback identified mistakes and offered task-specific corrections (RQ2a). Although the AI-generated feedback in this study provided significantly more guidance related to metacognitive processes (Nazaretsky et al. 2025), students did not fully recognise or value this aspect. Yet this type of feedback is critical for preparing students for lifelong learning in an AI-driven world. To address this gap, educational practitioners should foster student agency in the feedback process (Darvishi et al. 2024), emphasise to students the developmental value of metacognitive feedback for meaningful learning (Carless and Boud 2018), and cultivate both AI and feedback literacy (Carless and Winstone 2023; Long and Magerko 2020). AI literacy enables learners to critically engage with AI tools, understanding their strengths and limitations. Feedback literacy, in turn, equips students to interpret and apply feedback effectively, supporting continuous improvement and evaluative judgement (Bearman et al. 2024; Tai et al. 2018; Sadler 2014). These complementary literacies provide learners with the critical thinking and adaptability needed to navigate dynamic, AI-enhanced learning environments.

6.1.3 | Promoting Studies in Authentic Learning Environments

A distinctive aspect of our study is its use of authentic student responses, which likely elicited more genuine reactions by directly addressing their own performance. This approach

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contrasts with many comparative studies that investigate general students' perceptions and attitudes towards the adoption of AI-based educational solutions (Henderson et al. 2025; Nazaretsky, Mejia-Domenzain, et al. 2025) or rely on fictional or exemplar feedback to maintain experimental control (Abendschein et al. 2024; Kim et al. 2020; Radtke and Rummel 2025; Ruwe and Mayweg-Paus 2024). Other studies, particularly in the context of language education, involve students completing artificial essay-writing tasks and evaluating feedback from AI, instructors, or peers. These activities are often disconnected from students' actual learning contexts and routine assessment practises (Escalante et al. 2023; Meyer et al. 2024; Ruwe and Mayweg-Paus 2023; Tossell et al. 2024; Zeevy-Solovey 2024; Zou et al. 2025). However, using such controlled environments may significantly influence users' perceptions due to the human inclination to treat evidence differently depending on whether the issue is a personal matter or not (Nickerson 1998). For instance, teachers' perspectives on AI-based recommendations shifted dramatically, from predominantly favourable when applied to synthetic class data to highly critical when evaluating data from their own classrooms (Nazaretsky et al. 2021). Echoing (Abendschein et al. 2024; Zhang et al. 2025), we underscore the importance of conducting perception and bias-related studies within authentic learning environments, as they provide more realistic and contextually relevant insights into participants' perceptions, attitudes, and behaviours.

6.1.4 | Promoting Human-Mediated Automated Feedback Approaches (Human-In-The-Loop)

A promising approach to mitigating algorithm aversion and addressing potential AI inaccuracies is complementing human educators' input with AI-based systems (Henderson et al. 2025; Tossell et al. 2024; Zawiah et al. 2023; Zhang et al. 2025). In this hybrid model, AI systems generate initial feedback, which human experts review and validate to ensure high-quality, error-free, and pedagogically sound feedback before being shared with students. This approach leverages the speed and scalability of AI whilst incorporating the nuanced judgement and contextual understanding of human educators.

Notably, such hybrid systems can also address the social, emotional, and psychological dimensions of learning (Zhang et al. 2025). Research has shown that whilst students may view AI feedback as credible, they assign lower ratings to its goodwill when explicitly labelled as AI-generated (Abendschein et al. 2024). Our findings further emphasise that encouragement and appreciation hold value when delivered by humans, as they are perceived as genuine and meaningful. In contrast, the exact encouraging phrases from AI often feel hollow or mechanical, lacking the emotional authenticity that human interactions provide. Consequently, students are less likely to resonate with AI-delivered encouragement, perceiving it as generic rather than a sincere acknowledgment of their efforts. This highlights the intrinsic human need for recognition and emotional support, which AI, despite its capabilities, struggles to replicate effectively.

However, implementing the human-in-the-loop strategy necessitates new competencies for instructors, broadly categorised under the umbrella of 'teacher feedback literacy' (Boud and Molloy 2013; Carless and Winstone 2023). The evolving landscape of educational technology, particularly with freely available tools, demands that educators develop skills to effectively integrate automated feedback into their pedagogical practises (Buckingham Shum et al. 2023; Lazarus et al. 2024). This includes critically assessing and tailoring AI-generated feedback to meet the specific needs of individual students or student groups and comprehending the potential and limitations of AI tools, especially LLM, e.g., ChatGPT (Zhai 2023; Zhai and Nehm 2023). This comprehension includes critically evaluating the reliability and accuracy of the information provided by AI, being aware of possible model biases, and understanding how these factors might impact educational outcomes (Luckin et al. 2022). Recent studies have highlighted the importance of enhancing professional development programmes for teachers, focusing on co-designing AI-powered pedagogical solutions with teachers (Nazaretsky 2025) and increasing their epistemic understanding of AI (Nazaretsky, Ariely, et al. 2022; Kizilcec 2024).

7 | Limitations and Next Steps

Whilst this study involved 472 students in authentic learning environments, the sample may not fully capture the diversity of educational contexts and student demographics, particularly underrepresented groups such as non-binary students, thus limiting the generalizability of the findings. Additionally, the study did not account for individual differences in prior experience with Generative AI, students' knowledge levels, or the complexity of the assigned tasks. These factors may significantly influence how students perceive and evaluate AI-generated feedback and should be considered in future research. Furthermore, students' perceptions were captured at a single time point, offering only a snapshot of their attitudes. Longitudinal studies are needed to examine how repeated exposure to AI-generated feedback and growing familiarity with identifying such feedback may affect perceptions, trust, and acceptance over time. Finally, our study focused exclusively on students' perceptions. It did not investigate feedback uptake, that is, whether students are equally likely to act on feedback suggestions depending on the feedback's perceived source. Further large-scale empirical research is needed to examine how feedback source attribution impacts students' actual revision behaviours and learning outcomes. Additionally, beyond perceptions, future research should prioritise assessing hybrid approaches that incorporate human oversight (human-in-the-loop), particularly as the quality of generative AI feedback is expected to improve with advancements in its capabilities. Such investigations could provide valuable insights into how these increasingly sophisticated feedback mechanisms impact learning outcomes and how best to integrate them into instructional design to maximise their educational potential.

8 | Conclusions

Our study, involving 472 student responses in real learning environments, provided a nuanced understanding of how students evaluate human- and AI-generated feedback, offering insights beyond what synthetic scenarios might achieve. The findings

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revealed that students' evaluations were significantly influenced by their knowledge of the feedback provider's identity. Human feedback was rated slightly higher when the provider was disclosed, whilst AI feedback received lower ratings, particularly for *Genuineness*, where the decrease was significant. Students generally did not expect high-quality pedagogical feedback from Generative AI and were more likely to attribute superior feedback to the teaching team. This bias was reflected in Turing Test outcomes: students who rated AI feedback higher than human feedback in blind conditions often failed the test.

A key finding was the bidirectional influence between perceived feedback quality and perceived provider identity. Human feedback providers were consistently rated as more credible, and higher-quality feedback was overwhelmingly attributed to humans. This highlights the prevailing preference for human feedback in educational contexts and the challenges of integrating AI tools. Overall, our study sheds light on the complexities of student perceptions of AI feedback, offering essential guidance for the effective use of Generative AI in education.

Author Contributions

Tanya Nazaretsky: conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, software, visualization, writing – original draft, writing – review and editing. **Paola Mejia-Domenzain:** writing – review and editing, investigation. **Vinitra Swamy:** writing – review and editing, investigation. **Jibril Frej:** writing – review and editing, investigation. **Tanja Käser:** writing – review and editing, project administration, supervision, methodology, funding acquisition, conceptualization.

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Conflicts of Interest

The study was conducted with the approval of the Human Research Ethics Committee at EPFL (HREC 017-2023). The data from this study are not available online due to privacy issues. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

Research data are not shared.

Endnotes

[1] https://platform.openai.com/docs/guides/text.

[2] Parameters: model = gpt-4, top_p = 1.0, temperature = 0.8, frequency_penalty = 0.8, presence_penalty = 0.5.

[3] *stats* R package.

[4] *lme4* and *afex* R packages.

[5] *vif* function, *car* R package.

[6] https://translate.google.com/.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1.**

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Appendix A

TABLE A1 | Description of categories and subcategories emerged from the analysis of the responses.

Category	Description	Subcategories and Examples
Surface features		
Structure & grammar	Formatting consistency, flawless grammar.	The feedback text is well-structured (e.g., greeting, conclusion), nicely formatted (e.g., LaTeX, markdown, bullets, paragraphs, enumeration), with no spelling errors (e.g., typo, poor punctuation), and emojis (e.g., smiley faces), or uses more formal language (e.g., 'Vous' instead of 'Tu' or 'Cependant' instead of 'Mais' in French).
Clarity & readability	The extent to which feedback is concise and easy to understand.	Feedback text was characterised as direct, compact, understandable, coherent, short, focused (positive markers) or as complex, vague, unclear, longer, weird, incoherent, ambiguous, verbose, wordy, repetitive (negative markers).
Broader learning context	Learning and teaching routine habits, and pedagogical context.	The feedback referred to learning routine (e.g., online forum, class, and exam schedule or pedagogical team shortage in time/resources), made assumptions about learning and teaching habits (e.g., a student ran out of time, violated ethics by using ChatGPT), and was aligned with what happened in the class (e.g., absence of notation/methods/functions that are not part of the syllabus, specific professor's requirements and expectations).
Pedagogical features		
Objectivity	The extent to which feedback is precise, such as factual, personalised, and correct.	Some responses mentioned the feedback as precise, with no additional justification. In many other cases, though, students explained their opinions. The explanations were divided into three subcategories: (i) The feedback is true, honest, rational, and grounded in objective facts, free from subjective opinions or hallucinations. (ii) The feedback is personalised, specific, nuanced, tailored to the response, and based on student skills analysis. (iii) The feedback is based on a proper understanding of the response. It accurately identifies all the errors/flaws in the response and does not ignore any part of it. It does not ask to do something already done in the response.
Usefulness	The extent to which the feedback is applicable and helps to achieve the objectives of the assignment.	In some responses, the feedback was mentioned as helpful, instructive, useful, utile, or applicable (positive markers) vs. irrelevant, making insignificant, trivial, or unnecessary emphasis, too general or superficial to be helpful, not practical, or redundant (negative markers). In many other cases, the students provided more detailed explanations about feedback usefulness. These explanations were divided into two subcategories: (i) The feedback reflected on the student's current state of knowledge, e.g., pointing out what was done correctly and what was erroneous (positive markers) vs. not mentioning such information or just rephrasing the student's response with no arguments about its correctness (negative markers). (ii) The feedback provided advice on how to improve the answer, e.g., how to fix errors, what methods to apply, and how (positive markers) compared to the absence of such information (negative markers).
Genuineness	Human tone, empathy, encouragement.	The feedback text contains (i) Human emotion markers (e.g., friendly, natural, nice, kind, emotional, authentic, genuine, sincere, warm) vs. inhuman markers (e.g., cold, distant, robotic, unnatural, diplomatic, rude). (ii) Effective earned praise vs. over-positivity (e.g., over-encouraging, over-complementary, uncritical, too-enthusiastic, too-benevolent, too-kind).

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TABLE A2 | Student TT justification responses per category.

Category	Description	Student justification examples
Surface features		
Structure & grammar	Formatting consistency, flawless grammar.	'Ai feedback had more textbook english.'
		'There are two distinctive characteristics in the writing of AI. First, it uses enumeration, trying to follow a linear order in the feedback structure: "you started by", (then) "stated" (in the end) "and included". Also, AI often uses exclamation marks in the beginning and in the end (as in this case).'
		'Human feedback is not written in a formal and grammatically correct way.'
Clarity & readability	The extent to which feedback is concise and easy to understand.	'Human feedback is shorter and goes directly to the main point, real people don't have that much time and interest to give you so much "nice" but not that "useful & informative" sentences.'
		'AI feedback is shorter and too direct.'
		'AI feedback is more concise'
Broader learning context	Learning and teaching routine habits and pedagogical context.	'Human feedback [...] states the course slides seen in class.'
		'Human feedback contains the sentence' I guess you didn't have time to do it 'which leads me to think that it was written by a human'
		'AI feedback was [...] less linked to the vocabulary used in the lecture.'
		'AI did not know that we could use POWER instead of ^'
		'I don't think assistants have that much time to write such detailed and elaborate feedback for every student'
Pedagogical features		
Objectivity	The extent to which feedback is precise, meaning (i) factual, (ii) personalised, and (iii) correct.	'AI feedback is way less precise and does not give any information about my answer.'
		'AI feedback mentions unecessary code at the end of functions.py but there was nothing unecessary at the end of the file'
		'AI feedback is imprecise, and does not mention the problem set or any problems with our code except refactoring main.'
		'This mistake could've been missed by AI. This feedback have been written AI mostly because it didn't mention any mistake, but the other one did.'
		'AI feedback is more details but doesn't mention that it doesn't converge even it seems so'
		'Human feedback is more precise with information tailored to the case compared to AI Feedback.'
Usefulness	To what extent the feedback applicative and helps to achieve objectives of the assignment.	'AI feedback was also a bit less relevant because it focused on efficiency aspect which is less important in this context'
		'AI feedback gives rather general advice like splitting up code using functions'
		'AI feedback [...] goes through every step and explains in detail what i should've done'
		'Human feedback gives us hints but not the answers.'
		'Whilst the general conclusion of the two feedbacks are identical, human feedback gives a lot more detail on what *exactly* was done well and could be improved.'

(Continues)

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TABLE A2 | (Continued)

Category	Description	Student justification examples
Genuineness	Human tone, empathy, encouragement.	'AI feedback feels less authentic and colder.'
		'[it] is more human, meaning you can feel emotions in the words [...]'
		'Human feedback appears to be written by the teaching team because of the last sentence which encourages us to work correctly.'

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